

Here are some examples of composing functions together: If  $x = 3$  and  $f(x) = \pi x$ ,  $g(x) = \sin(x) + 1$ :

$$g(f(x)) = \sin(\pi \times 3) + 1 = 1 \quad (1.10)$$

$$f^3(x) = \pi^3 x \quad (1.11)$$

There is nothing much here, really, just evaluate the function from the innermost function to the outermost.